[Version B]

# Exam - 2025

## INFO9023: Machine Learning Systems Design

### Instructions

- Oral exam
- You will receive **1 use case** and a series of questions related to it
- Make sure to **thoroughly read** the use case description and each question
- **Motivate** your answers. Often the reason for making a specific design choice is as important as the choice itself.
- The total grade for the exam is **20 points**. Points are mentioned for each question

# Use case - Quality control with computer vision


**TechspaceMontef** is a company that builds automation software for advanced manufacturing plants. One of their clients operates several factories producing high-precision metal parts for the aerospace industry. Each part goes through a complex production line involving drilling, cutting, and polishing steps. Despite tight controls, microscopic surface defects - such as scratches, dents, and burns - occasionally occur and must be detected before going to the next stage of production or shipping.

Currently, inspection is done manually by quality control operators using magnifying cameras and checklists. The process is slow, expensive, and prone to human errors.

TechspaceMontef wants to deploy a **computer vision ML system** that automatically inspects parts. Cameras will be placed directly above conveyor belts at key stages of the manufacturing process. ML models must run using these images as input and output predictions on whether passing mechanical parts have any defects on it and - in case of defect - label which type of defect occurred.

The dataset includes **200.000 labeled images** from prior inspections, with bounding boxes and defect labels.

The constraints of different plants vary. Most will have fast moving conveyor belts. You can assume that each separate conveyor belt only processes a single type of part.

TechspaceMontef is setting up a team for building this ML application and you have been appointed as **Tech-Lead**. TechspaceMontef is aware of the importance of building a robust and scalable system. You are tasked with taking **key design decisions** regarding this application.

## Question 1 - Model training & optimisation

1.1. While designing your model, you want to be mindful of how many operations it will take to train it. What standard metric is used to calculate the computational cost of training a model? Cite two ways for calculating it.
*(1 point)*

1.2. You want to optimize the training of your model by parallelising it. Name and briefly explain the three types of model training parallelisation covered in this course..
*(2 points)*

1.3. You now want to simplify your model. Name and briefly explain three types of model complexity optimisation (or simplification) covered in this course.
*(2 points)*

## Question 2 - Model serving

2.1. Which part of your system will you run on premise or in the Cloud? Motivate your answer. Also give an example of a use case where the other deployment method would be applied.
*(3 points)*

2.2. Explain the concept of latency, throughput, auto-scaling and load balancing? How do they relate to each other, will some impact others?
*(2 points)*

*2.3.* Your clients are not happy with the latency of your system. Give the three types of optimisation possible to reduce latency <u>without changing your model</u> covered in this course?
*(2 points)*

## Question 3 - Model pipeline

3.1. What are the standard components (steps) of an ML pipeline? Which ones will you focus on in this project?
*(3 points)*

3.2. What types of triggers exist to automatically launch the ML pipeline? Which one would you recommend using in this use case?
*(1 point)*

## Question 4 - Monitoring

4.1. Explain the difference between resource and performance monitoring?
*(2 points)*

4.2. Give the four golden signals of resource monitoring?
*(2 points)*